Central Bank → Asset Prices

A Layered Decomposition from First Principles

A four-level structural framework for how monetary policy reaches asset prices. Each level adds a single layer of complexity: from a four-node flow at Level 1 to a closed-loop structural model at Level 4. Every node is re-explained at every level it appears, because the same node carries different meaning as the surrounding system grows richer.

Level 1 The simplest view: four nodes, one chain.

Level 2 Three policy instruments, term premium, three valuation drivers.

Level 3 The reaction function, expanded toolkit, intermediaries, FCI, feedback.

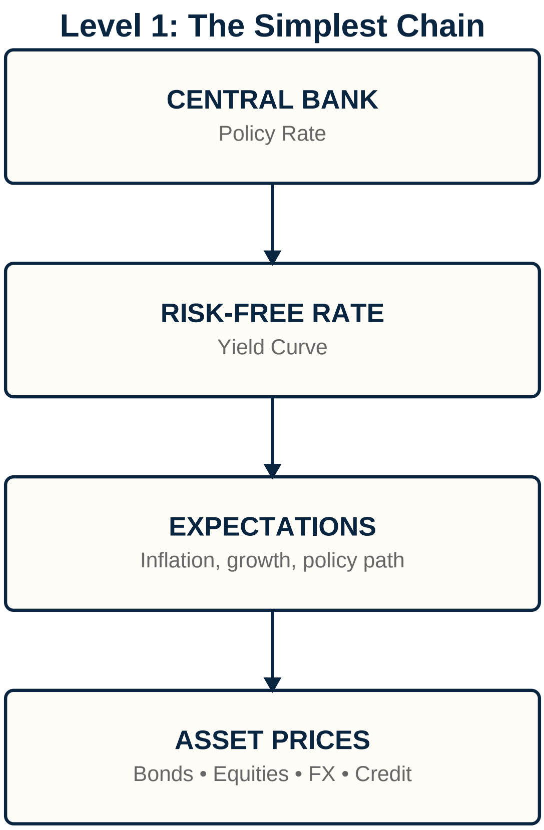
Level 4 Loss function, sub-instruments, five intermediaries, heterogeneous expectations, three terminal nodes.

Contents

Level 1

The Simplest View of a Central Bank

Level 1 is the minimum viable chart. It contains four nodes connected in a single chain: **Policy Rate** → **Yield Curve** → **Expectations** → **Asset Prices**. Every more complex view of monetary policy elaborates on this skeleton.



Each step transmits monetary policy to the next.

Node 1 — The Policy Rate

What it is

The policy rate is the interest rate the central bank charges or pays commercial banks for very short-term funds, typically overnight. In South Africa it is the repo rate; in the US the federal funds target range; in Brazil the SELIC. That is the entire definition: one overnight rate.

Why it is the only rate the bank directly controls

The central bank cannot decree what 10-year bonds yield, what mortgages cost, or what equities trade at. It can only set the price of overnight money between itself and the banking system. It controls this rate by being the monopoly supplier of bank reserves. That monopoly is the entire source of monetary policy power.

Why one overnight rate matters for everything else

Because every other interest rate in the economy is, in some sense, a string of expected overnight rates plus a premium. A 2-year yield is roughly the average overnight rate expected over the next two years. A 10-year yield is the same logic stretched further. Mortgage rates, corporate bond yields, deposit rates — all priced off this scaffolding. So when the central bank moves the overnight rate, the entire structure of interest rates re-prices.

> ***The simplest mental model:*** *The central bank sets one price — the cost of overnight money. Every other interest rate is built on top of that price. Move the foundation, and the whole structure shifts.*

Node 2 — The Risk-Free Rate / Yield Curve

What it is

The risk-free rate is the return on lending to the safest borrower in a given currency — usually the government — for a given length of time. The yield curve is that rate plotted across all maturities: 3 months, 2 years, 5 years, 10 years, 30 years. One line, showing the price of safe money at every horizon.

Why 'risk-free' and why it matters

It is risk-free because the government can, in its own currency, always pay you back — it prints the currency. The only compensation is for time, not default. Every other asset is then priced as 'risk-free rate plus something extra for the risk I'm taking.' Corporate bonds = risk-free + credit spread. Equities = risk-free + equity risk premium. Mortgages = risk-free + credit + prepayment. The risk-free curve is the foundation that every other asset price sits on top of.

How it connects to the policy rate

The very short end is essentially pinned to the policy rate. If the bank sets overnight money at 7%, no one will lend to the government for one month at 4% — they would just leave the money on deposit. The longer end is the market's average expectation of where the policy rate will be over that horizon. The policy rate anchors the front; expectations about future policy rates determine the rest.

> ***The simplest mental model:*** *The yield curve is the price of safe money at every length of time. The central bank pins down the start of the curve. The market draws the rest of the line based on where it thinks the central bank will go next.*

Node 3 — Expectations

What it is

Expectations are what market participants believe will happen next — to growth, inflation, policy rates, and corporate earnings. They are forecasts, held by millions of investors, aggregated into prices. Crucially, expectations are not facts — they are beliefs about the future that get priced today.

Why expectations sit between the curve and prices

The yield curve already contains expectations of future policy rates — that is what determines its shape beyond the front end. But asset prices need more than the path of rates. They need expectations about inflation (because real returns are what investors care about), growth (because it drives earnings and tax revenues), and future policy (because today's policy rate matters less than the path ahead). Expectations enrich the raw yield curve with the macro context needed to value anything beyond a government bond.

Why expectations are what the bank actually manages

A central bank meeting that does not change the policy rate can still move markets violently — if it changes expectations about the next meeting. The current policy rate affects very little economic activity directly. What matters is what borrowers and investors believe rates will be over the life of their decisions: the 30-year mortgage, the 10-year corporate investment, the 5-year bond purchase. Those are all priced off expected future rates. The bank's real job is shaping the expected path, not setting today's rate.

> ***The simplest mental model:*** *Today's policy rate matters less than what the market believes tomorrow's will be. Asset prices are bets on the future, so they live or die on expectations — and the central bank's job is to anchor those expectations.*

Node 4 — Asset Prices

What it is

Asset prices are what financial claims trade for in the market — bonds, equities, currencies, credit, real estate. They are the terminal node because everything the central bank does eventually shows up here.

The one equation behind every asset price

Price = Expected future cash flows ÷ Discount rate

The numerator is what you expect to receive (coupons, dividends, rents). The denominator is the rate at which you discount those future amounts back to today, because money tomorrow is worth less than money today. A bond, a stock, a building, a private equity stake — all priced by the same logic.

How the central bank reaches asset prices

Through the discount rate: by moving the yield curve, the bank directly changes what investors use to discount future cash flows. Higher rates → lower present value → lower prices. Through expectations of cash flows: by influencing growth and inflation expectations, the bank changes what investors think future cash flows will be. Both channels typically pull the same direction — a hike lowers the numerator and raises the denominator.

Why this is the terminal node

Asset prices are where monetary policy becomes visible. The policy rate is an abstraction; the yield curve is a chart on a screen; expectations live in surveys. Asset prices are the moment policy hits portfolios, balance sheets, and household wealth.

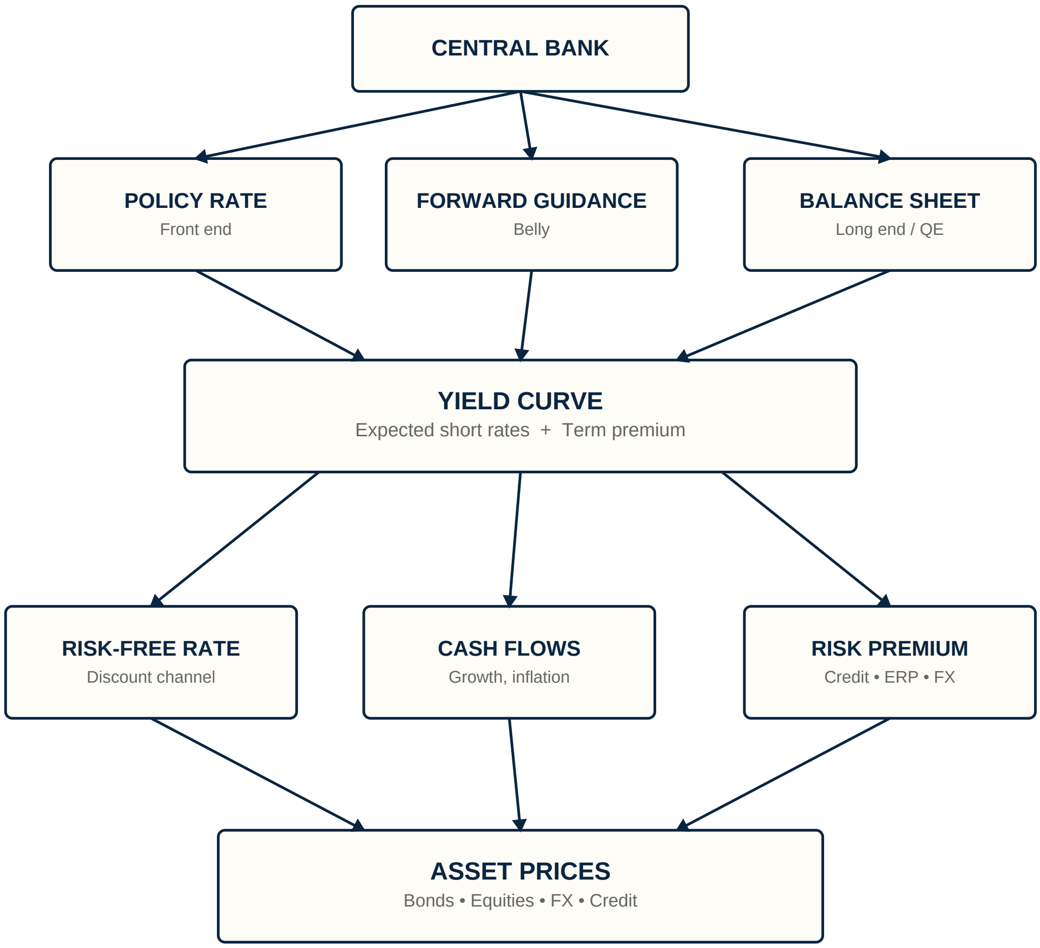
> ***The simplest mental model:*** *Every asset is a stream of expected future cash flows, discounted at some rate. The central bank moves both the cash flows and the discount rate. Asset prices are where that arithmetic plays out — and where policy stops being theory and starts being real.*

Level 2

Instruments, Curve Decomposition, Valuation Drivers

Level 2 unpacks each of the four Level 1 nodes. The single 'central bank' becomes three instruments. The yield curve splits into expected rates plus term premium. The discount rate splits into risk-free rate plus risk premium. Asset prices differentiate by class. The chain is the same — the resolution is finer.

Level 2: Instruments, Curve, Valuation Drivers



Each asset loads on the three drivers differently.

Govvies → rates dominant • Equities → all three • FX → relative

Node 1 — The Three Policy Instruments

At Level 1 the central bank had one tool. At Level 2 it has three, each targeting a different part of the yield curve.

The Policy Rate, in the Level 2 context

The policy rate at Level 2 is no longer the whole toolkit — it is one of three instruments. It moves the front end of the curve mechanically and immediately. Its limitation: it only directly controls overnight money. The rest of the curve is left to expectations and the balance sheet.

Forward Guidance

Forward guidance is the central bank talking about what it will do next. Words, when credible, move markets as effectively as rate changes — sometimes more so. If the bank says 'we expect to hold rates here for two years,' the market re-prices the entire path of expected short rates. The 2-year yield falls without the bank touching the policy rate. This works on the belly of the curve (1–5 years), where pricing depends on expected future policy. The mechanism is simple: long rates are averages of expected short rates. The limitation: it only works if the market believes you.

The Balance Sheet (QE / QT)

The balance sheet instrument is the central bank buying or selling bonds in the open market. Buying bonds (QE) injects reserves and removes duration from private hands; selling (QT) does the reverse. This works on the long end (5–30 years), where pricing depends less on expected short rates and more on the supply and demand of long-dated bonds. By absorbing duration, the bank pushes long-end yields down beyond what rate cuts alone would achieve. The limitation: the effect is harder to measure, slower, and politically fraught.

Why three instruments instead of one

Policy rate → front end • Forward guidance → belly • Balance sheet → long end

This is the key Level 2 insight. The central bank used to be a one-instrument actor. Since 2008 it operates across the entire curve — which is why the transmission diagram had to expand.

> ***The simplest mental model:*** *The central bank has three levers, each pulling on a different part of the yield curve. The rate moves the front, guidance moves the middle, the balance sheet moves the back. Together they let the bank shape the whole curve, not just its starting point.*

Node 2 — The Yield Curve Decomposed

At Level 1, the yield curve was 'the price of safe money at every horizon.' At Level 2 we open it up and look at what's inside each yield.

The two components of any long yield

Long yield = Expected average short rate + Term premium

Expected average short rate is what you would earn rolling overnight money for the full maturity. If the market expects the policy rate to average 6% over ten years, that is the first component.

Term premium is the extra compensation investors demand for locking money up rather than rolling overnight. It is the price of bearing duration risk — the risk that rates move against you while you are locked in.

A 10-year yield of 9% might be 6% expected short rates + 3% term premium. Or it might be 8% expected + 1% premium. Same yield, very different meaning.

Why this decomposition matters

The same headline move in the 10-year yield can mean opposite things depending on which component drove it. A rise driven by expected short rates is hawkish — markets think the bank will hike more. A rise driven by term premium is something else entirely — fiscal worry, inflation uncertainty, QT — and policy stance has not changed. A risk analyst looking only at the headline yield misses this.

How the three instruments map to the two components

Policy rate, forward guidance → expected short rates | Balance sheet → term premium

The first two instruments work through expectations. The balance sheet works through the term premium. That is why post-2008 central banks needed the third instrument — they had run out of room to push expectations lower at the zero bound, so they had to attack the term premium directly.

> ***The simplest mental model:*** *Every long yield is two things stacked: what the market expects rates to average, and what it charges extra for the risk of being wrong. The central bank's three instruments target these two components — and knowing which is moving tells you what's actually happening.*

Node 3 — The Three Valuation Drivers

At Level 1, asset prices came from one equation: cash flows ÷ discount rate. At Level 2, we split the discount rate into two pieces, giving three drivers.

Price = Expected cash flows ÷ (Risk-free rate + Risk premium)

The three components

Expected cash flows — what investors expect to receive. Coupons, dividends, rents, earnings. Driven by growth and inflation expectations.

Risk-free rate — what you would earn on a safe government bond of similar maturity. This is the yield curve from Node 2 flowing in.

Risk premium — the extra compensation investors demand for holding a risky asset instead of a safe one. Equity risk premium, credit spread, FX risk premium, illiquidity premium — all sit here.

Why splitting the discount rate matters

Because a hike does not always raise the full discount rate by the same amount. The risk-free portion rises mechanically with policy. But the risk premium can move in either direction. A credible hike can compress risk premia — the market trusts the bank to control inflation, so it demands less compensation for uncertainty. A non-credible hike can widen risk premia — the market sees the bank panicking, fears recession or default, and demands more compensation. The same 100bp policy move can produce different total discount rate changes. This is exactly the asymmetry that matters for emerging markets.

How the central bank reaches each driver

Risk-free rate — directly, through the yield curve. **Cash flows** — indirectly, by influencing growth and inflation expectations. **Risk premium** — indirectly, through credibility and financial conditions. The first channel is mechanical and fast. The second is slow and works through the real economy. The third is psychological and depends entirely on whether the market trusts the bank.

> ***The simplest mental model:*** *Every asset price is a tug-of-war between three forces: what you'll be paid, what safe assets pay, and what extra you demand for the risk. The central bank pulls on all three — but only the first one is mechanical. The other two depend on whether the market believes what the bank is doing will work.*

Node 4 — How the Drivers Hit Different Assets

Same three drivers — risk-free rate, expected cash flows, risk premium — but each asset class loads on them differently. That is why the same policy shock produces different responses across equities, fixed income, and FX.

Equities

Price = Expected earnings ÷ (Risk-free rate + Equity risk premium)

All three drivers are live and roughly equally weighted. The risk-free rate hits long-duration equities (tech, growth) hardest because their cash flows sit far in the future. Cash flows respond to growth expectations — a hike that slows the economy lowers earnings forecasts. The equity risk premium widens in stress. A typical hike: all three usually pull the same direction, which is why equities are sensitive to monetary policy.

Fixed Income

Government bonds are almost pure risk-free rate plus term premium. Cash flows are fixed and certain. There is no credit risk premium in own currency. So a govvie price moves almost entirely on the discount rate channel. **Credit** (corporate, EM hard currency) adds a credit spread. Cash flows are still mostly fixed (coupons), but now there is default risk.

$$\text{Govvie } \Delta P \approx -\text{Duration} \times (\Delta\text{Risk-free rate} + \Delta\text{Term premium})$$

$$\text{Credit } \Delta P \approx -\text{Duration} \times (\Delta\text{Risk-free rate} + \Delta\text{Credit spread})$$

Note that the duration framework and the discounted cash flow framework are the same equation viewed differently. Duration is the first derivative of the DCF equation with respect to yield. DCF tells you the level; duration tells you the sensitivity. For practitioners — risk attribution, scenario analysis, P&L; decomposition — duration is the working language. For cross-asset comparison, DCF is the common framework since it generalises beyond fixed income.

FX

FX is the odd one out — it is not a discounted cash flow asset. It is the relative price of two currencies, so it depends on the difference between two countries' rates and risk premia.

FX move ≈ Relative interest rate differential + Relative risk premium

A hike that raises domestic rates above foreign makes the currency more attractive (carry). But if the hike signals panic or worsens the growth outlook, the currency risk premium widens and capital flows out. FX has its own credibility asymmetry: a credible hike sees rate differential dominate and the currency strengthens; a non-credible hike sees risk premium dominate and the currency weakens. This is why EM FX is the cleanest read on central bank credibility.

Cross-asset summary

Asset	Risk-free rate	Cash flows	Risk premium
Equities	Strong	Strong	Strong
Govvies	Dominant	Fixed	Term premium only
Credit	Strong	Fixed	Credit spread
FX	Differential	n/a	Relative premium

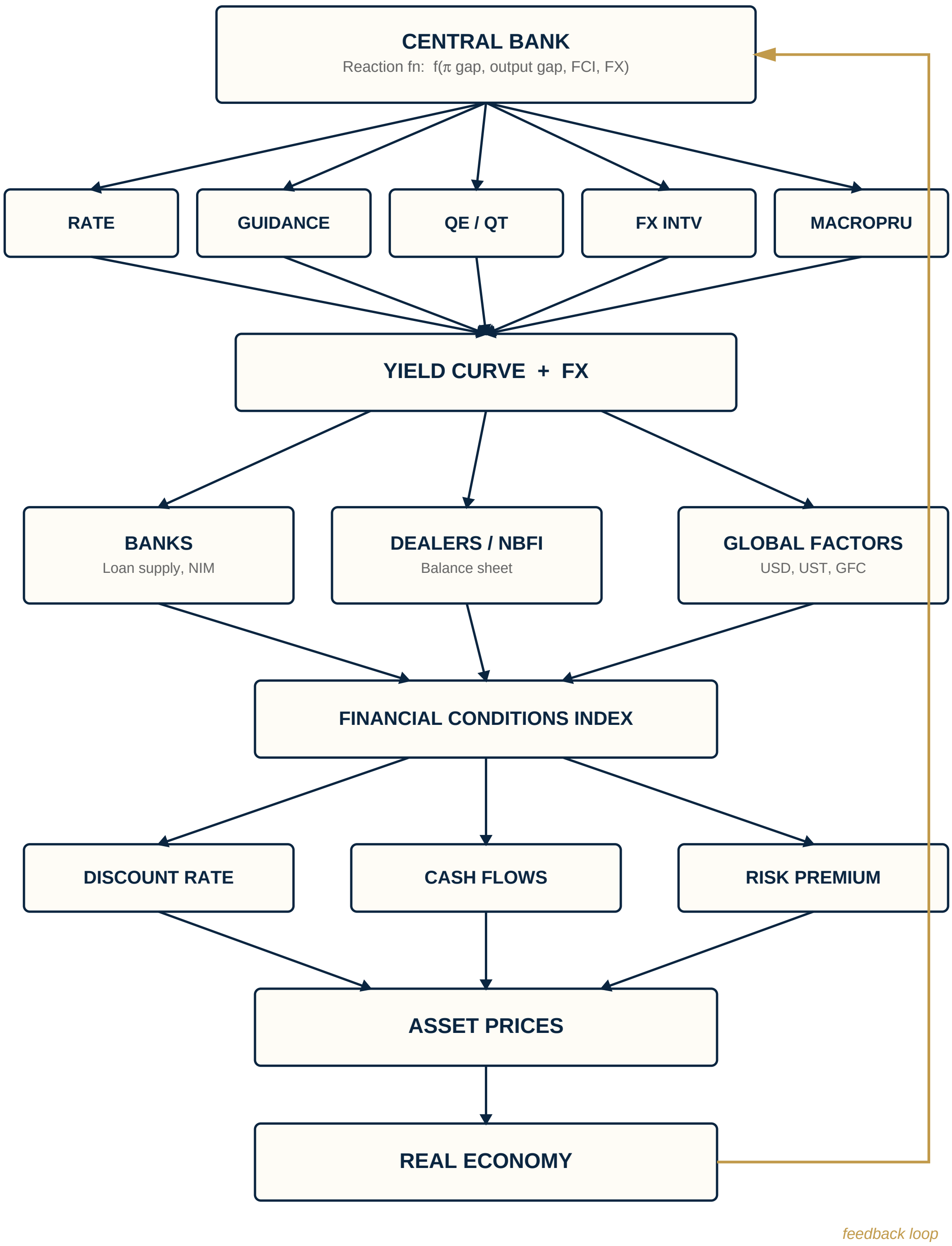
> ***The simplest mental model:*** *Every asset feels the same three forces, but with different sensitivities. Govvies are almost pure rates. Equities feel all three. Credit adds default risk. FX is the relative version of the whole exercise — and the cleanest read on central bank credibility.*

Level 3

Reaction Function, Intermediaries, Feedback Loop

Level 3 is the conceptually biggest jump. The central bank stops being an exogenous shock generator and becomes endogenous — it follows a reaction function. New institutions sit between the curve and asset prices. And the whole system closes into a loop, with asset prices feeding the real economy and the real economy feeding the bank's next decision.

Level 3: Reaction Function, Intermediaries, Feedback



Node 1 — The Reaction Function

At Levels 1 and 2, the central bank simply 'acted' and the system responded. At Level 3, the bank becomes endogenous: it responds to the state of the world according to a known rule.

The textbook Taylor rule

$$\textit{Policy rate} = \textit{Neutral rate} + \alpha \times (\textit{Inflation gap}) + \beta \times (\textit{Output gap})$$

If inflation is above target, hike. If output is below potential, cut. The α and β coefficients are the bank's type — a hawk has high α, a dove has high β.

Why this matters for asset prices

Once the bank has a known reaction function, the data itself becomes the policy signal. A high CPI print is not just news about inflation — it is news about the next rate decision. Markets re-price the entire yield curve before the bank meets, because they can compute what the bank will do. This is why CPI prints often move bonds more than actual rate decisions: the decision is already priced in by the time it arrives. The surprise in the data is what does the work.

The EM extension

Most EM central banks have reaction functions that include more than inflation and output:

Policy rate = f(Inflation gap, Output gap, FX, Financial conditions)

A weak currency forces hikes even when inflation and growth would suggest holding. This is the 'fear of floating' extension (Calvo–Reinhart). It is why EM curves are more sensitive to FX moves than DM curves — the market knows the bank will respond to currency.

Why credibility lives here

Credibility is the market's confidence in the stability of the reaction function. A credible bank's coefficients are believed to be stable. A non-credible bank's coefficients might shift — toward fiscal accommodation, political pressure, or panic. When credibility is high, the market does most of the work for the bank, pricing in the appropriate response to data automatically. When credibility is low, the bank has to do everything manually — and even then, the market may not believe it.

> ***The simplest mental model:*** *The central bank is no longer just acting — it's responding to the economy according to a known rule. Once the market understands the rule, the data becomes the policy signal. Credibility is the market's confidence that the rule won't change.*

Node 2 — The Expanded Instrument Set

At Level 2, the bank had three instruments. At Level 3, two more appear — FX intervention and macroprudential tools — because real central banks (especially EM) operate across more than just the yield curve.

FX Intervention

FX intervention is the central bank buying or selling its own currency. Buying foreign currency (selling its own) weakens the domestic currency; the reverse strengthens it. The policy rate works on domestic interest rates; FX intervention works directly on the exchange rate. Sterilised intervention offsets the liquidity effect so the policy rate is unaffected — which means it is a standalone instrument independent of monetary stance. EM banks use it heavily because FX is in their reaction function.

Macroprudential Tools

Macroprudential policy targets financial stability directly, rather than working through the macro economy. The toolkit includes the countercyclical capital buffer (CCyB), loan-to-value (LTV) caps, debt-service-to-income (DSTI) caps, and reserve requirements. These target who gets credit and on what terms, not the overall price of money. You can have loose monetary policy and tight macroprudential policy simultaneously — exactly what many DM banks did post-2008.

Why two mandates require two instrument sets

This is the key Level 3 insight, formalised by Tinbergen: you need at least as many instruments as objectives. A bank with two objectives — price stability and financial stability — needs at least two instruments. The policy rate alone cannot hit both. Monetary instruments target price stability; FX intervention targets the exchange rate; macroprudential targets financial stability. Each mandate gets its own toolkit.

> ***The simplest mental model:*** *A modern central bank has two jobs — keep prices stable, keep the financial system safe — and it can't do both with one tool. The instrument set splits accordingly: monetary tools for the economy, macroprudential for the banking system, FX intervention for the currency.*

Node 3 — The Intermediary Layer

This is the core of Level 3. At Levels 1 and 2, monetary policy flowed straight from the yield curve to valuation drivers. At Level 3, institutions sit in between — and they have constraints that distort transmission.

Banks

Banks decide how much of any rate change to pass through. Three things shape pass-through: **loan supply** (a hike that hits bank capital can shrink lending more than the rate alone implies — Bernanke–Blinder bank lending channel); **deposit beta** (how much of a rate change banks pass to depositors — a low deposit beta widens NIM but weakens transmission); **net interest margin** (banks protect NIM, so they do not always pass changes through symmetrically). The central bank can hike 100bp, but if banks raise lending rates by only 60bp, the effective tightening is smaller than the policy move suggests.

Dealers and Asset Managers

The post-2008 addition. Dealers (who make markets in bonds) and asset managers (who hold them) are constrained by regulatory capital (Basel III, SLR, G-SIB surcharges), risk limits (VaR caps), and funding (repo, dollar funding, prime brokerage). When balance sheets are constrained, they cannot absorb risk even when prices look attractive. The He–Krishnamurthy and Adrian–Etula insight: risk premia are a function of intermediary balance sheet health, not just policy stance. This is why March 2020 happened — the Fed was easing aggressively, but dealer balance sheets were so constrained that Treasuries became illiquid and yields rose during a flight to safety.

Global Factors

For an EM bank — and increasingly for everyone — domestic policy is not the only force on domestic asset prices. Global factors arrive through the dollar (DXY), US Treasury yields, and global risk appetite (VIX, Global Financial Cycle). The Miranda-Agrippino–Rey insight: there is a global financial cycle driven largely by the Fed and global risk appetite. EM asset prices comove with this cycle whether the domestic central bank likes it or not. The classical trilemma becomes a dilemma — even with floating FX, you do not have full monetary independence.

Why the intermediary layer matters

The same policy shock produces different asset price responses depending on the state of the intermediary layer. A 100bp hike when banks are well-capitalised and dealers have ample balance sheet → smooth transmission. The same hike when banks are stretched and dealers are constrained → amplified response, possible market dislocation, risk premia spike. This is why linear models of policy transmission fail in stress periods. The intermediary state is a non-linear conditioning variable on transmission strength.

> ***The simplest mental model:*** *Between the central bank and asset prices sit institutions with their own constraints. Banks decide how much policy to pass through. Dealers decide how much risk they can carry. Global factors arrive whether you want them or not. The same policy shock has different effects depending on whether these intermediaries are healthy or stretched.*

Node 4 — The FCI and the Feedback Loop

The Financial Conditions Index

An FCI is a single number that bundles the state of financial markets into one transmission gauge. The major ones — Goldman's GS-FCI, the Chicago Fed NFCI, the Fed's FCI-G — typically combine short rates (policy stance), long rates (yield curve), credit spreads (corporate borrowing cost), equity prices (wealth effect, cost of capital), and the exchange rate (trade-weighted). The weights are calibrated so that a one-unit FCI move corresponds to a roughly equivalent impact on GDP a few quarters out.

Why the FCI sits where it does in the chart

Because the FCI is what actually transmits to the real economy — not the policy rate alone. A 100bp hike offset by tighter credit spreads, falling equities, and a stronger currency produces a much larger FCI tightening than the rate move alone implies. Conversely, a hike accompanied by easing credit spreads and rising equities (because the market sees the bank as credible and growth as resilient) produces almost no FCI tightening — the rate hike 'didn't take.' This is exactly why central banks watch FCIs closely. The policy rate is the instrument; the FCI is what does the work.

The feedback loop

This is the Level 3 closure. Asset prices are no longer terminal — they feed back into the system through the real economy. **Asset prices → real economy:** the wealth effect on consumption, cost of capital on investment, exchange rate on inflation, bank lending on credit supply. **Real economy → central bank:** the reaction function from Node 1 closes the loop. Once asset prices have moved the real economy, the bank's reaction function tells you what it does next. The system is now fully circular.

Stabilising vs destabilising loops

The geometry of the loop determines whether shocks dampen or amplify. **Stabilising (DM, credible bank):** Hike → FCI tightens → growth slows → inflation falls → bank pauses. Each round of feedback dampens the original shock. **Destabilising (EM, low credibility):** Hike → FX weakens (markets price more hikes) → inflation rises → bank must hike more → growth collapses → fiscal stress → FX weakens further. Each round amplifies. This is the classic EM crisis dynamic. The same instrument can be stabilising or destabilising depending entirely on whether the loop dampens or amplifies — and what determines that is the bank's credibility, the intermediary state, and fiscal space.

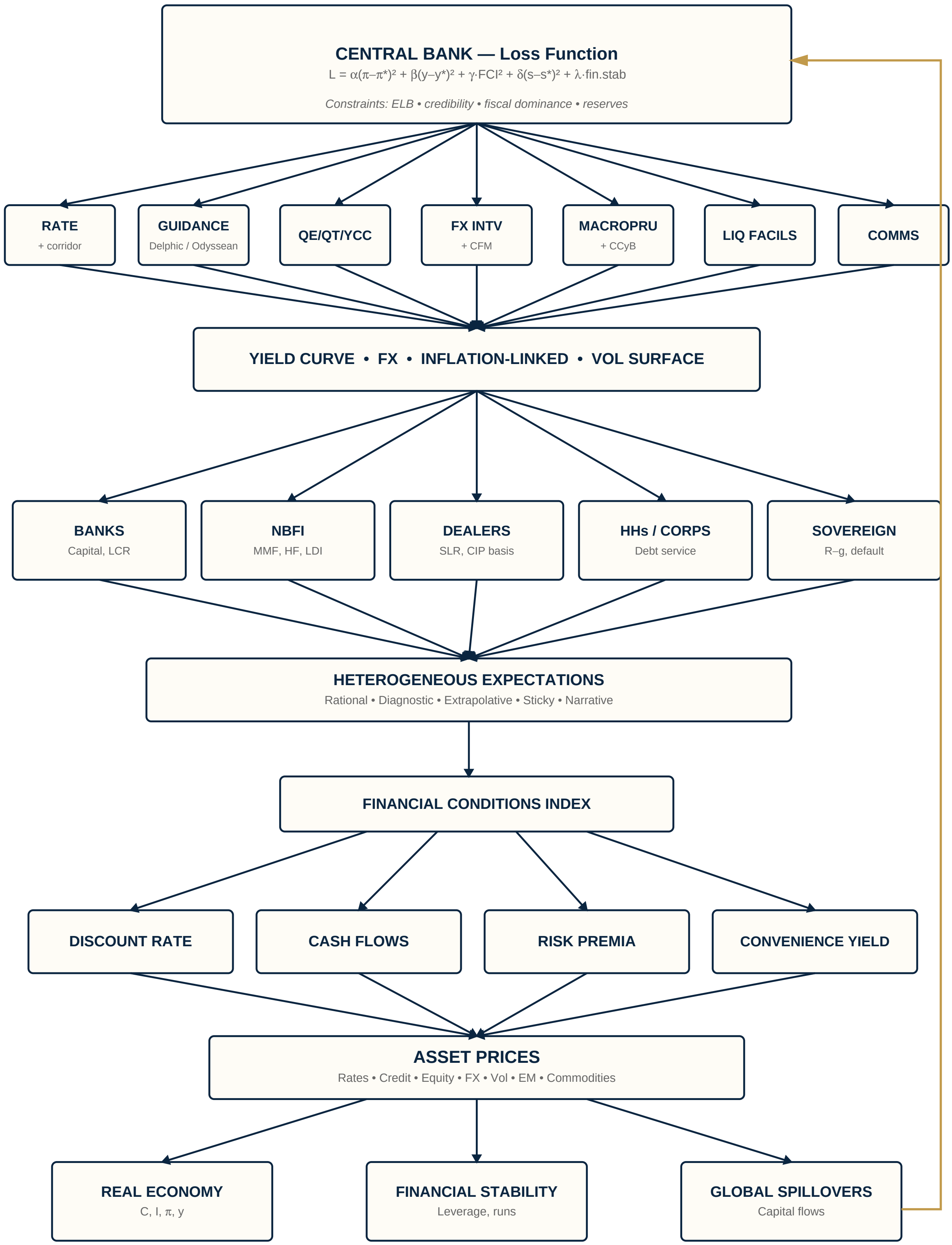
> ***The simplest mental model:*** *The FCI is the real transmission gauge — not the rate. Asset prices feed back into the economy, which feeds back into the central bank's next decision. The system is a loop, and whether shocks dampen or amplify depends on credibility, intermediary health, and fiscal space.*

Level 4

Structural Model: Loss Function, Sub-Instruments, Five Intermediaries, Heterogeneous Expectations, Three Terminal Nodes

Level 4 stops being a flowchart and becomes a structural model. The reaction function is shown to be the first-order condition of an underlying loss minimisation problem. Each instrument splits into sub-instruments. The intermediary layer expands from three to five distinct nodes with named frictions. Expectations become heterogeneous, and a fourth valuation driver — the convenience yield — appears. Asset prices feed into three parallel terminal nodes, all with bidirectional global spillovers.

Level 4: Structural Model



Node 1 — The Loss Function and Constraints

At Level 3, the reaction function gave us a rule the bank follows. At Level 4, we ask the deeper question: where does that rule come from? The answer is that the bank is minimising a loss function subject to constraints. The reaction function is just the first-order condition of the optimisation.

The loss function

$$L = \alpha(\pi - \pi^*)^2 + \beta(y - y^*)^2 + \gamma \cdot FCI^2 + \delta(s - s^*)^2 + \lambda \cdot (\text{financial stability})$$

Each term is a squared deviation from a target. Squaring penalises both directions and captures the convexity that large deviations hurt disproportionately more than small ones — a 5% inflation overshoot is far more than five times as bad as a 1% overshoot. The coefficients α, β, γ, δ, λ are how much the bank cares about each deviation.

The coefficients are the bank's 'type'

Every central bank has the same functional form, but different weights. The weights *are* the bank's identity. A hawk has high α (cares deeply about inflation); a dove has high β (tolerates inflation to support employment); an EM bank with FX anchor has high δ; a post-2008 bank has positive λ; a fragile EM has high δ and high λ. Two banks facing identical conditions choose different policy rates because their loss functions weight deviations differently. This is why 'what would the Fed do?' is not the same question as 'what would the SARB do?'

The constraints

The loss function is minimised subject to constraints — and these matter as much as the objective:

- **Effective lower bound (ELB):** you cannot cut below ~0%. Conventional policy becomes non-linear at the bottom.
- **Credibility stock:** if the market does not believe you, your instruments do not work. Credibility takes years to build, can be destroyed in one meeting.
- **Fiscal dominance:** when government debt is high enough, hiking worsens fiscal sustainability — the Sargent-Wallace 'unpleasant arithmetic.'
- **FX reserves:** for EM banks intervening in FX, you can only sell what you have.
- **Political independence:** formally independent banks still face pressure that constrains action.

Why this matters for asset prices

Three reasons. **First**, knowing the loss function and constraints lets you solve for the optimal policy path given current data — what professional central bank watchers do when they estimate α, β, γ, δ, λ. **Second**, it explains regime shifts: when the weights change (e.g. λ moves from zero to positive after a financial crisis), the entire reaction function shifts. **Third**, it makes credibility quantifiable. Credibility is the market's confidence that the coefficients are stable. A non-credible bank is one whose α might collapse under political pressure, whose δ might suddenly dominate in a crisis, whose λ might appear out of nowhere. The asset price response to any shock depends on whether the market trusts the coefficients to hold.

> ***The simplest mental model:*** *The central bank is solving an optimisation problem: minimise the squared deviations of inflation, output, financial conditions, the currency, and stability — weighted by how much it cares about each — subject to hard constraints. The reaction function is just the answer to that problem. Different weights produce different banks. Changing weights produce regime shifts. Mispricing the weights produces alpha.*

Node 2 — The Fully Unpacked Instrument Set

At Level 3, the bank had five instruments. At Level 4, each splits further — because the details of how each instrument is deployed matter for transmission. We move from 'what tool' to 'how it's used.'

Policy Rate + Corridor Mechanics

The policy rate is no longer a single number. It is a corridor: **interest on reserves (IOR)** at the floor, **policy rate** at the mid, **discount window/standing facility** at the ceiling. The width of the corridor determines how cleanly the policy rate transmits. A narrow corridor pins overnight rates tightly; a wide corridor lets market rates drift, weakening transmission. Post-2008, central banks added overnight reverse repo (ON-RRP) and standing repo facilities (SRF) to manage the floor and ceiling more precisely — sub-instruments that did not exist before.

Forward Guidance: Delphic vs Odyssean

Campbell et al. split forward guidance into two distinct types. **Delphic**: the bank shares its forecast ('we expect rates at 5% next year given our outlook'). This is information about the economy, not a commitment. **Odyssean**: the bank commits to a path ('we will hold rates here through 2025 even if inflation rises'). This is a binding pledge. Odyssean guidance moves markets more if credible, because it overrides future data. The trouble: it only works if the market believes the bank will honour it under future inflation surprises, which most banks will not. The Fed's 2020-2021 inflation guidance is the cautionary tale.

Balance Sheet: QE / QT / YCC

The balance sheet splits into three modes. **QE** buys bonds, expands balance sheet, pushes down term premium. **QT** lets bonds roll off, raises term premium. **Yield curve control (YCC)** commits to buy whatever quantity is needed to hold a specific yield at a specific maturity. The structural difference: QE/QT target quantities; YCC targets prices. YCC is more powerful when credible — the market does the work of holding the yield — but more dangerous when challenged, because defending the peg can require unlimited buying.

FX Intervention + Capital Flow Management

Two sub-instruments. **Sterilised FX intervention** buys/sells FX without changing domestic liquidity — pure FX channel. **Capital flow management (CFM)** imposes controls on cross-border flows: taxes on inflows (Brazil's IOF), unremunerated reserve requirements on portfolio inflows (Chile's encaje), outright outflow restrictions in crisis. CFM directly targets the quantity of capital crossing the border. The IMF formally legitimised CFM in 2012 as a valid EM tool under specific conditions.

Macroprudential + CCyB

Static macropru sets fixed rules: Basel III ratios, LTV caps, DSTI caps. **Time-varying macropru** is the active piece — the countercyclical capital buffer (CCyB) and dynamic provisioning tighten in booms and loosen in busts. This lets the bank lean against credit cycles without touching rates.

Liquidity Facilities

A separate instrument set that operates in stress: the **discount window** for standard collateralised lending; **FX swap lines** (Fed lines to ECB, BoE, BoJ, SNB are the dollar liquidity backstop for the global system); the **standing repo facility (SRF)** for committed repo lending against Treasuries; and crisis facilities (TALF, PDCF, BTFP) created on the fly. These target funding directly, not rates or asset prices. Dormant in normal times; the most important tool in crisis.

Communications as a Standalone Instrument

Beyond forward guidance: the statement (parsed line-by-line), the press conference (tone, body language, ad-libs), the dot plot/SEP (distribution of member forecasts — dispersion matters as much as median), and inter-meeting speeches. A dovish press conference after a hawkish hike can ease financial conditions despite tighter rates. Communication is sometimes the dominant transmission channel.

The Tinbergen problem at Level 4

With this many instruments and only a handful of objectives, instruments can work at cross-purposes. A hike + dovish guidance is an ambiguous net stance. QT + cautious communications is ambiguous. Tighter macropru + looser monetary has different distributional effects than tighter monetary alone. The market has to read the vector of all instruments simultaneously to infer the bank's actual stance. This is why parsing FOMC days is genuinely hard.

> ***The simplest mental model:*** *Each instrument from Level 3 has a more granular form. The corridor matters, not just the headline rate. Guidance has two flavours with different binding power. Balance sheet operations target prices or quantities. FX intervention separates from capital controls. Liquidity facilities are dormant until they aren't. Communications move markets independently of decisions. Reading policy means reading the whole vector.*

Node 3 — The Intermediary Layer Expanded

At Level 3, the intermediary layer had three sub-nodes: banks, dealers/asset managers, global factors. At Level 4, it expands to five distinct nodes, each with specific frictions that distort transmission.

Banks (expanded)

The Level 3 view (loan supply, deposit beta, NIM) gets sharper. **Capital ratios** — Basel III CET1 minimums constrain risk-bearing; hit the minimum, lending shrinks. **Liquidity coverage ratio (LCR)** — banks must hold high-quality liquid assets, affecting which bonds they buy. **Asset-liability mismatch** — banks borrow short and lend long; SVB 2023 was this friction breaking under fast hikes. Banks transmit policy only as far as their balance sheet allows.

NBFI (Non-Bank Financial Intermediaries)

The genuinely new Level 4 node. Post-2008, risk migrated out of banks into **money market funds (MMFs)** (short-term funding, runs in stress — March 2020), **hedge funds** (leveraged long-short, basis trades), **insurers and pensions** (long-duration liabilities, forced sellers in extreme moves — UK LDI 2022), and **private credit/private equity** (opaque leverage, mark-to-market lags). Post-2008 regulation tightened on banks but largely missed NBFI. Most recent crisis episodes — 2019 repo, March 2020 Treasury dysfunction, 2022 LDI, 2023 regional banks — were NBFI-driven, not bank-driven. The transmission map from textbooks is increasingly incomplete.

Dealers and Market-Makers

Level 3 covered balance sheet capacity. Level 4 names the specific frictions. **SLR (supplementary leverage ratio)** limits dealer Treasury holdings. **G-SIB surcharges** add capital for globally systemic banks; year-end window dressing distorts repo markets. **CIP basis** — covered interest parity used to hold; post-2008 it does not, because dealers cannot arbitrage it without expanding balance sheet. The basis *is* the friction, priced in basis points. **Dollar funding** — non-US institutions need dollars; cross-currency basis swaps reveal the stress. Dealer balance sheet is the plumbing of fixed income markets. When constrained, prices stop reflecting fundamentals and start reflecting capacity.

Households and Corporates

A new Level 4 node — the demand side of credit, not just the supply side. **Debt service ratios** — how much of income goes to debt repayment; high ratios mean rate hikes hit harder. **Wealth effect** — asset price moves change consumption directly. **Heterogeneous MPCs** — high-debt households cut spending sharply on hikes; low-debt households barely respond; aggregate response depends on the distribution. The same policy stance has very different effects depending on the balance sheet composition of the private sector.

Sovereign / Fiscal

The most important EM addition. The sovereign is now inside the intermediary system. **Debt-to-GDP** — high debt means hikes worsen fiscal sustainability. **R – g dynamics** — when interest rates exceed growth, debt compounds; tipping point for fiscal dominance. **Issuance schedule** — heavy supply weeks raise term premium independently of policy. **Default risk** — sovereign CDS prices the tail. In EM, the sovereign is not a passive participant but a price-maker. A hike that makes the fiscal arithmetic untenable can raise yields rather than lower them, because credit risk dominates rate risk.

The five-node insight

The intermediary layer is no longer 'banks + dealers + global.' It is a system of five distinct intermediary types, each with its own frictions and failure modes. Different shocks stress different nodes. **Slow hikes** hit banks (NIM, capital). **Fast hikes** hit dealers (balance sheet capacity) and NBFI (funding runs). **Crisis** hits all five with contagion. **EM stress** hits the sovereign first, then the rest.

> ***The simplest mental model:*** *Five intermediary types, each with specific frictions: banks (capital), NBFI (runs), dealers (balance sheet), households/corporates (debt service), sovereign (fiscal arithmetic). The same policy shock stresses different nodes depending on speed, size, and context. Transmission strength depends on which nodes are healthy and which are stretched.*

Node 4 — Heterogeneous Expectations and the Convenience Yield

At Level 3, expectations were a single node — a representative agent forming rational forecasts. At Level 4, two refinements: agents are heterogeneous, and a fourth valuation driver appears.

Heterogeneous Expectations

Replace the single rational agent with a mixture of belief types: a **rational core** (uses all available information, unbiased forecasts); **diagnostic expectations** (Bordalo–Gennaioli–Shleifer — overweight recent news); **extrapolative** agents (project recent trends forward); **sticky information** (Mankiw–Reis — update slowly, anchored to old beliefs); and **narrative-driven** agents (Shiller — beliefs spread like contagion through stories). With heterogeneous agents, predictable forecast errors exist. The market over-extrapolates recent inflation, then under-reacts to the bank's response, then over-corrects. This generates momentum, reversal, and convexity — features rational expectations cannot produce.

The credibility implication. Diagnostic agents over-update on recent inflation prints. A bank that lets inflation drift faces convex re-anchoring losses — small drift is cheap to fix; large drift is exponentially expensive because diagnostic agents have re-anchored their priors. This is the structural argument for hitting inflation targets early and hard.

Convenience Yield (the fourth valuation driver)

Levels 1–3 had three valuation drivers: discount rate, cash flows, risk premium. Level 4 adds a fourth. The **convenience yield** (Krishnamurthy–Vissing-Jorgensen) is the non-pecuniary benefit of holding safe assets — Treasuries, bunds, dollar cash. Investors accept a lower yield to hold them because of **collateral value** (accepted in repo, derivatives, settlements), **liquidity** (sellable at scale), **regulatory utility** (LCR, capital relief), and **flight-to-quality demand** (in stress, everyone wants them simultaneously).

Yield = Risk-free rate + Term premium − Convenience yield

The convenience yield subtracts from yield because investors pay (in foregone yield) for the convenience.

Why convenience yield is a separate driver

Because it moves independently of the other three. A flight-to-quality episode can spike the convenience yield by 50bp without any change in expected rates, growth, or risk premia. QE compresses convenience yield by removing safe assets from supply. Debt ceiling episodes spike it by threatening the safe-asset status of T-bills.

For EM: this is critical. Local-currency EM bonds trade against a global convenience yield they do not control. When the global convenience yield spikes (US Treasuries become more 'convenient'), capital rotates out of EM into the safe-asset pool. The EM yield rises not because EM fundamentals deteriorated, but because the relative convenience of US assets jumped. This is the cleanest framework for understanding the 2013 taper tantrum, March 2020, and any 'dollar shortage' episode.

How the two refinements interact

Heterogeneous expectations × convenience yield produces non-linear stress dynamics. A small shock hits diagnostic agents, who over-extrapolate. They demand more safe assets, spiking the convenience yield. The convenience yield spike tightens global financial conditions. Tighter conditions hit EM through the intermediary layer. Sticky-info agents in EM are slow to update, so positioning unwinds violently when they do. This is the structural backbone of why crises feel non-linear and self-reinforcing.

> ***The simplest mental model:*** *Markets are not a single rational agent — they're a mixture of belief types that produce momentum, reversal, and non-linear stress. Safe assets carry a convenience yield that subtracts from yield and spikes in flight-to-quality. EM gets hit through this channel even when domestic fundamentals are unchanged. Together, heterogeneous expectations and convenience yield explain why crises feel non-linear and why credibility is structural rather than scalar.*

Node 5 — Three Terminal Nodes and Bidirectional Spillovers

The final Level 4 node closes the entire system. At Level 3, the chart had one terminal node (the real economy) feeding back to the central bank. At Level 4, there are three terminal nodes, each feeding back independently.

Three terminal nodes

Asset prices now feed into three parallel destinations.

- **Real economy** — consumption, investment, inflation, output. Feeds back via the standard reaction function (α, β coefficients).
- **Financial stability** — leverage, maturity mismatch, fire-sale externalities, runs, contagion. Feeds back via the λ coefficient.
- **Global spillovers** — capital flows, FX, foreign asset prices. Feeds back via foreign reaction functions affecting domestic conditions.

The real economy and financial system can move in opposite directions. Asset prices can be supporting consumption (good for real economy) while leverage builds dangerously (bad for stability). The bank can no longer collapse outcomes into a single output — it must track all three.

Why financial stability is a separate mandate

The 2008 crisis showed that price stability does not guarantee financial stability. The Great Moderation (1985-2007) had stable inflation and stable output, while leverage built to crisis levels underneath. The textbook objective function was hitting its targets while the system broke. Central banks need a separate mandate, with separate instruments (macroprudential), tracking separate outputs (leverage, mismatch, concentration). Monetary policy and macroprudential can pull in opposite directions. Loose money + tight macropru is the post-2008 DM playbook. Tight money + loose macropru can occur in EM stress.

Bidirectional global spillovers

At Level 3, global factors were exogenous. At Level 4, spillovers are bidirectional. **Inbound (Level 3 view):** Fed policy → global FCI → your domestic asset prices; dollar strength → your currency, your inflation; global risk appetite → your capital flows. **Outbound (Level 4 addition):** your policy aggregates with peers to move global EM risk premia; a coordination failure across EMs (2013 taper tantrum) amplifies the shock for everyone; large EMs (China, India, Brazil) genuinely move global commodity prices, growth, and FCI.

The small open economy assumption breaks for large EMs. They are simultaneously takers of global conditions and contributors to them. Policy decisions need to anticipate not just domestic response but how peers respond and how the aggregate returns.

Closing the full loop

The complete Level 4 system loops as follows: **Central bank (loss function) → instruments → yield curve → intermediaries (5 nodes) → FCI → valuation drivers (4) → asset prices → three terminal nodes → reaction function → back to central bank.** Each terminal node feeds back through a different coefficient in the loss function. The real economy feeds α and β. Financial stability feeds λ. Global spillovers feed δ and the constraint set.

This is why modern central banking is genuinely hard. The bank is no longer hitting one target with one instrument — it is optimising a five-coefficient loss function across multiple instruments, watching three distinct terminal nodes, accounting for foreign banks doing the same, all under hard constraints.

> ***The simplest mental model:*** *The system has three outputs, not one. The real economy and financial stability can diverge — that's why the bank needs separate mandates and instruments. Global spillovers are bidirectional — your policy affects peers, theirs affects you, and large EMs are part of the global cycle they're navigating. The full loop is five coefficients, multiple instruments, three terminal nodes, all simultaneous.*

End of decomposition. Four levels, sixteen nodes, every node re-explained at every level it appears. The chart has progressed from a four-node flow to a closed-loop structural model. The natural next step is to apply this framework to specific shocks — a hawkish Fed surprise, an EM credibility break, a fiscal dominance episode — and watch how the cross-asset response decomposes through each level.